

Multiple Choice (10 marks)

1. Use the van der Waals parameters for chlorine to calculate approximate values of the radius of a Cl_2 molecule regarded as a sphere.
 - (a) 0.185 nm
 - (b) 0.110 nm
 - (c) 0.125 nm
 - (d) 0.200 nm
 - (e) 0.095 nm
2. An electron was accelerated through a potential difference of 1.00 ± 0.01 kV. What is the uncertainty of the position of the electron along its line of flight?
 - (a) $6.75 \times 10^{-10} \text{m}$
 - (a) $8.22 \times 10^{-11} \text{m}$
 - (a) $2.15 \times 10^{-12} \text{m}$
 - (a) $1.885 \times 10^7 \text{ m s}^{-1}$
 - (a) $1.22 \times 10^{-9} \text{m}$
3. Express the van der Waals parameters $b = 0.0226 \text{ dm}^3 \text{ mol}^{-1}$ in SI base units.
 - (a) $22.6 \times 10^{\{-3\}} \text{ m}^3 \text{ mol}^{-1}$
 - (b) $0.0226 \text{ m}^3 \text{ mol}^{-1}$
 - (c) $2.26 \times 10^{\{-6\}} \text{ m}^3 \text{ mol}^{-1}$
 - (d) $226 \times 10^{\{-6\}} \text{ m}^3 \text{ mol}^{-1}$
 - (e) $2.26 \times 10^{\{-2\}} \text{ m}^3 \text{ mol}^{-1}$
4. Of what importance are proteins to biological systems?
 - (a) Proteins determine the rate of photosynthesis in plants.
 - (b) Proteins are used for insulation in cold environments
 - (c) Proteins are the main constituent of cell walls in bacteria.
 - (d) Proteins are responsible for coloration of organisms
 - (e) Proteins serve as structural material and biological regulators.
5. The weight percent of sodium hydroxide dissolved in water is 50%. What is the mole fraction of sodium hydroxide?

- (a) 0.311
- (b) 0.5
- (c) 1.00
- (d) 0.05
- (e) 0.10

True / False (10 marks)

Answer True or False

6. True or False: The pH of the solution formed by adding 40 ml of 0.10 M NaOH to 10 ml of 0.45 M HCl is 2.
7. True or False: A 3.00 liter solution of 0.500 N sulfuric acid contains 102.5 grams of H_2SO_4 .
8. True or False: Differences in electronegativity between hydrogen and deuterium account for the shift in infrared absorption frequency of DCl and HCl.
9. True or False: The maximum kinetic energy of the α -particles emitted in the radioactive decay of ${}^6\text{He}$ is 2.01 MeV.
10. True or False: In a solution of 98 g H_2SO_4 dissolved in 162 g H_2O , the mole fraction of H_2SO_4 is 0.1 and that of H_2O is 0.9.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the work done when a gas in a 2.0-liter container at 2.4 atmospheres expands against an external pressure of 0.80 atmosphere. Discuss the implications of this process.
12. Discuss the relationship between pH and chemical shifts in the ${}^{31}\text{P}$ NMR spectrum of sodium phosphate. Specifically, analyze the expected chemical shift when the pH equals the pK_a of H_2PO_4 .

End of Paper